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SENSOR SOAP DISPENSER

Abstract

A soap dispenser including a dispensing spout configured to be supported above a mounting deck. A sensor is supported by the dispensing spout and is in electrical communication with a controller supported below the mounting deck. The controller includes a normal operating mode and an anti-clog operating mode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application claims priority to U.S. Provisional Patent Application Ser. No. 63/557,036, filed Feb. 23, 2024, the disclosure of which is expressly incorporated herein by reference.

BACKGROUND AND SUMMARY OF THE DISCLOSURE

[0002] The present invention relates generally to a soap dispenser and, more particularly, to a sink deck mounted, electronic sensor soap dispenser including a below deck controller providing for normal and anti-clog modes of operation.

[0003] Electronic soap dispensers including sensors for hand-free operation are known in the art. Such electronic soap dispensers may include proximity sensors to detect the presence of a user and dispense soap in response thereto.

[0004] According to an illustrative embodiment of the present disclosure, a soap dispenser includes a dispensing spout configured to be supported above a mounting deck and including an outlet. A mounting shank is operably coupled to the dispensing spout and is configured to extend downwardly from the dispensing spout through the mounting deck. A soap reservoir is fluidly coupled to the dispensing spout and is configured to hold liquid soap below the mounting deck. A pump is fluidly coupled to the soap reservoir for pumping liquid soap to the outlet of the dispensing spout. The pump is configured to be supported below the mounting deck. A sensor is supported by the dispensing spout and is configured to provide an outlet signal when a user is within a detection area near the dispensing spout. A controller is in electrical communication with the sensor and is positioned below the mounting deck. The controller is configured to receive the output signal from the sensor and activate the pump to dispense liquid soap in response to the output signal.

[0005] According to a further illustrative embodiment of the present disclosure, a soap dispenser includes a dispensing spout configured to be supported above a mounting deck and including an outlet. A mounting shank is operably coupled to the dispensing spout and is configured to extend downwardly from the dispensing spout through the mounting deck. A soap reservoir is fluidly coupled to the dispensing spout and is configured to hold liquid soap below the mounting deck. A pump is fluidly coupled to the soap reservoir for pumping liquid soap to the outlet of the dispensing spout. A sensor is supported by the dispensing spout and is configured to provide an output signal when a user is within a detection area near the dispensing spout. A controller is in electrical communication with the sensor and is configured to receive the output signal from the sensor and activate the pump to dispense liquid soap in response to the output signal. The controller illustratively includes a normal operating mode and an anti-clog operating mode. The controller selecting one of the normal operating mode and the anti-clog operating mode based upon the position of the user within the detection area and the duration of the user within the detection area.

[0006] According to another illustrative embodiment of the present disclosure, a soap dispenser includes a dispensing spout configured to be supported above the mounting deck and including an outlet. A mounting shank is operably coupled to the dispensing spout and is configured to extend downwardly from the dispensing spout through the mounting deck. A soap reservoir is fluidly coupled to the dispensing spout and is configured to hold liquid soap below the mounting deck. A pump is fluidly coupled to the soap reservoir for pumping liquid soap to the outlet of the dispensing spout. A sensor is supported by the dispensing spout and is configured to provide an output signal when a user is within a detection area near the dispensing spout. A controller is in electrical communication with the sensor and is configured to receive the output signal from the sensor and to activate the pump to dispense liquid soap in response to the output signal. The controller includes a normal operating mode and an anti-clog operating mode. The controller selects the operating mode based upon the position of the user within the detection area and the duration of the user within the detection area. The detection area includes a first detection limit and a second detection limit, the second detection limit being closer to the sensor then the first detection limit. The first detection limit defines a normal operating mode and the second detection limit defines the anti-clog operating mode. When the controller is in the normal operating mode, it will cause the pump to dispense liquid soap for a first duration. When the controller is in the anti-clog operating

mode, it will cause the pump to dispense liquid soap for a second duration, the second duration being greater than the first duration.
[0007] Additional features and advantages of the present invention will become apparent to those skilled in the art upon consideration of the following detailed description of the illustrative embodiment exemplifying the best mode of carrying out the invention as presently perceived.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] A detailed description of the drawings particularly refers to the accompanying figures, in which:
[0009] FIG. **1** is a front perspective view of an illustrative sensor soap dispenser of the present disclosure shown mounted on a sink deck adjacent to a kitchen faucet;
[0010] FIG. **2** is a rear perspective view of the illustrative sensor soap dispenser of FIG. **1**;
[0011] FIG. **3** is a front perspective view of the dispensing spout of the illustrative sensor soap dispenser of FIG. **1**;
[0012] FIG. **4** is a front exploded perspective view of the dispensing spout of FIG. **3**;
[0013] FIG. **5** is a rear exploded perspective view of the dispensing spout of FIG. **3**;
[0014] FIG. **6** is a block diagram of components of the illustrative sensor soap dispenser of FIG. **1**;
[0015] FIG. **7** is a side elevational view of the illustrative sensor soap dispenser of FIG. **1**, showing a normal mode of operation;
[0016] FIG. **8** is a side elevational view of the illustrative sensor soap dispenser of FIG. **7**, showing an anti-clog mode of operation;
[0017] FIG. **9** is a perspective view of the illustrative sensor soap dispenser of FIG. **3**, showing a soap reservoir refill operation; and
[0018] FIG. **10** is a detailed perspective view showing coupling of the dispensing spout and the mounting shank of the sensor soap dispenser of FIG. **9**.
DETAILED DESCRIPTION OF THE DRAWINGS
[0019] For the purposes of promoting and understanding the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, which are described herein. The embodiments disclosed herein are not intended to be exhaustive or to limit the invention to the precise form disclosed. Rather, the embodiments are chosen and described so that others skilled in the art may utilize their teachings. Therefore, no limitation of the scope of the claimed invention is thereby intended. The present invention includes any alterations and further modifications of the illustrated devices and described methods and further applications of principles in the invention which would normally occur to one skilled in the art to which the invention relates.
[0020] Referring initially to FIG. **1**, an illustrative electronic soap dispenser **10** is shown coupled to a mounting or sink deck **12** defining a sink basin **14** and positioned adjacent to a conventional kitchen faucet **16**. The kitchen faucet **16** illustratively includes a water outlet **18** to dispense water into the sink basin **14**. The water outlet **18** is illustratively defined by a sprayhead **20** supported by a spout **22**. The spout **22** may be supported for swiveling movement by a hub **24**. The water outlet **18** is in fluid communication with a conventional mixing valve **26** via an outlet tube **28** and controlled via a input or control handle **30**. Water is supplied to the mixing valve **26** via a hot water source (e.g., hot water stop) **32** and a cold water source (e.g., cold water stop) **34** via a hot water inlet line **36** and cold water inlet line **38**, respectively.
[0021] The spout **22** of the illustrative kitchen faucet **16** is a spring spout including a spring **40** receiving the outlet tube **28**. An arm **42** releasably couples the sprayhead **20** to the spout **22**. While the illustrative kitchen faucet **16** is a spring spout faucet, other types of faucets may be substituted

therefor.

[0022] The soap dispenser **10** illustratively includes a dispensing head or spout **50** supported above the mounting deck **12** and including a liquid soap outlet **52**. A mounting shank **54** illustratively extends downwardly from the dispensing spout **50** through the mounting deck **12**.

[0023] With reference to FIGS. **1** and **2**, the illustrative mounting shank **54** is a hollow tubular member including a cylindrical side wall **56** extending vertically along a longitudinal axis **57** between a first, upper end **58** and a second, lower end **60**. The side wall **56** includes external threads **62**. A threaded nut **64** is configured to cooperate with the external threads **62** of the mounting shank **54** to couple the dispensing spout **50** to the mounting deck **12**.

[0024] A liquid soap reservoir **66** is operably coupled to the dispensing spout **50** via the mounting shank **54**. A pump **70** is configured to pull liquid soap from the soap reservoir **66** to the outlet **52** of the dispensing spout **50**. More particularly, an outlet tube **72** couples the pump **70** to the outlet **52**. A soap refill passageway **73** extends within the mounting shank **54** from the upper end **58** axially downwardly to the reservoir **66** (FIG. **10**). The reservoir **66** is illustratively defined by a plastic container **75** including an open upper end **74** in fluid communication with the lower end **60** of the mounting shank **54**. A supply tube **76** illustratively extends between the pump **70** and a lower end **78** of the soap reservoir **66**.

[0025] With reference to FIGS. **3-5**, the illustrative dispensing spout **50** includes a body **80** having a first or lower shell **82** and a second or upper shell **84** defining a passageway **86** receiving a portion of the outlet tube **72**. The lower shell **82** illustratively includes a mounting flange **87**. The outlet tube **72** illustratively extends between a first end **88** and a second end **90**. The first end **88** is fluidly coupled to the pump **70**, while the second end **90** is fluidly coupled to a nozzle **92** defining the soap outlet **52** (FIGS. **2** and **4**). As shown in FIGS. **1**, **2** and **9**, the outlet tube **72** extends upwardly from the pump **70** external to the mounting shank **54**, then through the sidewall **56** of the mounting shank **54**, through the upper end **58** of the mounting shank **54** and within the passageway **86** of the dispensing spout **50**.

[0026] A proximity sensor **93** is supported by the dispensing spout **50** and is configured to provide an output signal when a user is within a detection area or zone **96** near the dispensing spout **50**. Illustratively, the proximity sensor **93** may comprise a first or long-range infrared (IR) sensor **94***a*, and a second or near-range infrared (IR) sensor **94***b*. Each infrared sensor **94***a*, **94***b* illustratively includes an emitter **98***a*, **98***b* to emit infrared light, and a receiver **100***a*, **100***b* configured to receive reflected infrared light from the emitter **98***a*, **98***b*. The emitter **98***a*, **98***b* and the receiver **100***a*, **100***b* are received within a housing **101**. The infrared sensors **94***a*, **94***b* of the proximity sensor **93** may determine distance and duration of a user (e.g., hands) within the detection area **96**.

[0027] While a pair of infrared sensors **94***a*, **94***b* are illustrated, it may be appreciated that the proximity sensor **93** may be defined by any number of infrared sensors (e.g., a single infrared sensor). Alternatively, other types of proximity sensors (e.g., inductive, capacitive, ultrasonic, etc.) may be substituted therefor.

[0028] The detection area **96** illustratively includes a first detection zone or limit **102** and a second detection zone or limit **104**. Illustratively, the first detection zone **102** is further from the dispensing spout **50** (and hence the proximity sensor **93**) than the second detection zone **104** (FIGS. **3**, **7** and **8**). More particularly, the first detection zone **102** extends a first distance **103** from the proximity sensor **93**, and the second detection zone **104** extends a second distance **105** from the proximity sensor **93**. As further detailed herein, the first distance **103** is greater than the second distance **105**.

[0029] With reference to FIGS. **2** and **6**, a controller **106** is in electrical communication with the sensor **93** and is configured to receive the output signal from the infrared sensors **94***a*, **94***b* and activate the pump **70** to dispense liquid soap in response to the output signal. Illustratively, both the controller **106** and the pump **70** are located within a control box **108** positioned below the sink deck **12**. An electrical cable **110** provides an electrical connection between the sensor **94** and the controller **106**. A quick release coupler **112** (such as a socket **114** and a plug **116**) may be provided in the cable **110** to assist in installation and servicing.

[0030] Illustratively, a battery **118** is also supported below the sink deck **12** and is in electrical communication with the controller **106** via an electrical cable **120**. A quick release coupler **122** (such as a socket **124** and a plug **126**) may be provided in the cable **120** to assist in installation and servicing. The battery **118** may include batteries (e.g., four AA batteries) received within a battery box **128**.

[0031] With reference to FIGS. **9** and **10**, a mount **130** is supported by the upper end **58** of the mounting shank **54** and defines the soap refill passageway **73** (for supplying liquid soap to the reservoir **48** as shown by arrow **132** in FIG. **9**). A releasable coupler **134** is defined between a lower end **136** of the dispensing spout **50** and the upper end **58** of the mounting shank **54**. Illustratively, the coupler **134** includes first tabs **138** on the dispensing spout **50**, and cooperating second tabs **140** on the mounting shank **54**. The tabs **138** and **140** define a bayonet type coupler to couple the dispensing spout **50** to the mounting shank **54**. A user may access the soap refill passageway **73** by rotating the spout **50** (as shown by arrow **139**) and then axially lifting the spout **50** vertically (as shown by arrow **141**).

[0032] A light indicator **142** is illustratively supported by the dispensing spout **50** and is in electrical communication with the controller **106**. The electrical cable **110** may provide an electrical connection between the light indicator **142** and the controller **106**. The light indicator **142** may comprise a light emitting diode (LED) and is configured to provide an indication of the operating mode of the controller **106**. The light indicator **142** may also provide an indication of other operating conditions, such as low battery power, low liquid soap in the reservoir **66** and/or service required.

[0033] With further reference to FIGS. **3**, **7** and **8**, the detection area **96** includes the first detection zone or limit **102** and the second detection zone or limit **104**. The second detection zone **104** is closer to the sensor **93** supported by the dispensing spout **50** than the first detection zone **102**. Illustratively, distance **103** of the first detection zone **102** is approximately 3 inches +/−0.45 inches from the sensor **93**, and the distance **105** of the second detection zone **104** is approximately 1 inch +/−0.45 inches from the sensor **93**. The first detection zone **102** defines a normal operating mode, and the second detection zone **104** defines an anti-clog operating mode.

[0034] In the illustrative normal operating mode, when the first infrared sensor **94***a* detects a user's hands within the first detection zone **102** for a first predetermined sensing time period (approximately 0.25 seconds), then the output signal from the sensor **94***a* will cause the controller **106** to operate the pump **70** for dispensing liquid soap for a first duration or predetermined dispensing time period (approximately 3 seconds) until activated again.

[0035] In the illustrative anti-clog operating mode, when the second infrared sensor **94***b* detects a user's hands within the second detection zone **104** for a second predetermined sensing time period (approximately 5 seconds), then the output signal from the sensor **94***b* will cause the controller **106** to operate the pump **70** for dispensing liquid soap for a second duration or predetermined dispensing time period (approximately 10 seconds). This extended dispensing time helps prevent liquid soap from being blocked in the outlet tube **72**. During dispensing in the anti-clog operating mode, the light indicator **142** will illustratively flash (illustratively 0.3 seconds every 1 second) as a mode indication to the user.

[0036] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the spirit and scope of the invention as described and defined in the following claims.

Claims

1. A soap dispenser comprising: a dispensing spout configured to be supported above a mounting deck and including an outlet; a mounting shank operably coupled to the dispensing spout and configured to extend downwardly from the dispensing spout through the mounting deck; a soap reservoir fluidly coupled to the dispensing spout and configured to hold liquid soap below the mounting deck; a pump fluidly coupled to the soap reservoir for pumping liquid soap to the outlet of the dispensing spout, the pump configured to be supported below the mounting deck; a sensor supported by the dispensing spout and configured to provide an output signal when a user is within a detection area near the dispensing spout; and a controller in electrical communication with the sensor and positioned below the mounting deck, the controller configured to receive the output signal from the sensor and to activate the pump to dispense liquid soap in response to the output signal.

2. The soap dispenser of claim 1, further comprising: a soap supply tube external to the mounting shank and fluidly coupled between the pump and the dispensing spout; and a soap refill passageway defined within the mounting shank and including an inlet positioned above the mounting deck and an outlet positioned below the inlet and in fluid communication with the soap reservoir.

3. The soap dispenser of claim 2, further comprising: a mount supported by the mounting shank and defining the inlet; and a releasable coupler defined between a lower end of the dispensing spout and the mount of the mounting shank.

4. The soap dispenser of claim 1, further comprising a battery in electrical communication with the controller and configured to be supported below the mounting deck.

5. The soap dispenser of claim 1, further comprising a light indicator supported by the dispensing spout and in electrical communication with the controller.

6. The soap dispenser of claim 1, wherein the sensor is an infrared sensor including an emitter and a receiver.

7. The soap dispenser of claim 1, wherein the controller includes a normal operating mode and an anti-clog operating mode, the controller selecting the operating mode based upon the position of the user within the detection area and the duration of the user within the detection area.

8. The soap dispenser of claim 7, wherein the detection area includes a first detection zone and a second detection zone, the second detection zone closer to the sensor than the first detection zone, the first detection zone defining the normal operating mode and the second detection zone defining the anti-clog operating mode.

9. The soap dispenser of claim 8, wherein the first detection zone is defined by a long-range infrared sensor, and the second dictation zone is defined by a near-range infrared sensor.

10. The soap dispenser of claim 7, wherein: the controller in the normal operating mode will cause the pump to dispense liquid soap for a first duration; the controller in the anti-clog operating mode will cause the pump to dispense liquid soap for a second duration; and the second duration is greater than the first duration.

11. The soap dispenser of claim 1, wherein the pump and the controller are received within a control box under the mounting deck.

12. A soap dispenser comprising: a dispensing spout configured to be supported above a mounting deck and including an outlet; a soap reservoir fluidly coupled to the dispensing spout and configured to hold liquid soap; a pump fluidly coupled to the soap reservoir for pumping liquid soap to the outlet of the dispensing spout; a sensor supported by the dispensing spout and configured to provide an output signal when a user is within a detection area near the dispensing spout; a controller in electrical communication with the sensor, the controller configured to receive the output signal from the sensor and to activate the pump to dispense liquid soap in response to the output signal; and wherein the controller includes a normal operating mode and an anti-clog operating mode, the controller selecting one of the normal operating mode and the anti-clog operating mode based upon the position of the user within the detection area and the duration of the user within the detection area.

13. The soap dispenser of claim 12, wherein the detection area includes a first detection zone and a second detection zone, the second detection zone closer to the sensor than the first detection zone, the first detection zone defining the normal operating mode and the second detection zone defining the anti-clog operating mode.

14. The soap dispenser of claim 13, wherein the first detection zone is defined by a long-range infrared sensor, and the second dictation zone is defined by a near-range infrared sensor.

15. The soap dispenser of claim 12, wherein: the controller in the normal operating mode will cause the pump to dispense liquid soap for a first duration; the controller in the anti-clog operating mode will cause the pump to dispense liquid soap for a second duration; and the second duration is greater than the first duration.

16. The soap dispenser of claim 12, further comprising: a mounting shank operably coupled to the dispensing spout; a soap supply tube external to the mounting shank and fluidly coupled between the pump and the dispensing spout; and a soap refill passageway defined within the mounting shank and including an inlet positioned above the mounting deck and an outlet positioned below the inlet and in fluid communication with the soap reservoir.

17. The soap dispenser of claim 12, further comprising a light indicator supported by the dispensing spout and in electrical communication with the controller, the light indicator configured to provide an indication of the operating mode of the controller.

18. The soap dispenser of claim 12, wherein the sensor is an infrared sensor including an emitter and a receiver.

19. A soap dispenser comprising: a dispensing spout configured to be supported above a mounting deck and including an outlet; a mounting shank operably coupled to the dispensing spout and configured to extend downwardly from the dispensing spout through the mounting deck; a soap reservoir fluidly coupled to the dispensing spout and configured to hold liquid soap below the mounting deck; a pump fluidly coupled to the soap reservoir for pumping liquid soap to the outlet of the dispensing spout; a sensor supported by the dispensing spout and configured to provide an output signal when a user is within a detection area near the dispensing spout; a controller in electrical communication with the sensor, the controller configured to receive the output signal from the sensor and to activate the pump to dispense liquid soap in response to the output signal; wherein the controller includes a normal operating mode and an anti-clog operating mode, the controller selecting the operating mode based upon the position of the user within the detection area and the duration of the user within the detection area; wherein the detection area includes a first detection limit and a second detection limit, the second detection limit closer to the sensor than the first detection limit, the first detection limit defining the normal operating mode and the second detection limit defining the anti-clog operating mode; wherein the controller in the normal operating mode will cause the pump to dispense liquid soap for a first duration; and wherein the controller in the anti-clog operating mode will cause the pump to dispense liquid soap for a second duration, the second duration being greater than the first duration.

20. The soap dispenser of claim 19, further comprising: a soap supply tube external to the mounting shank and fluidly coupled between the pump and the dispensing spout; and a soap refill passageway defined within the mounting shank and including an inlet positioned above the mounting deck and an outlet positioned below the inlet and in fluid communication with the soap reservoir.

21. The soap dispenser of claim 20, further comprising: a mount supported by the mounting shank and defining the inlet; and a releasable coupler defined between a lower end of the dispensing spout and the mount of the mounting shank.

22. The soap dispenser of claim 19, further comprising a light indicator supported by the dispensing spout and in electrical communication with the controller, the light indicator configured to provide an indication of the operating mode of the controller.

23. The soap dispenser of claim 19, wherein the sensor is an infrared sensor including an emitter and a receiver.

24. The soap dispenser of claim 19, wherein the first detection limit is defined by a long-range infrared sensor, and the second dictation limit is defined by a near-range infrared sensor.